

# BRYAN WEI XUAN CHEONG

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## EDUCATION

**Imperial College London** – MSc in Business Analytics (Distinction, Grade: 78%) Aug 2024 - Aug 2025  
*Modules:* Generative AI & LLMs, Statistics & Econometrics, Optimisation & Decision Models, Financial Analytics, Text Analytics, Digital Marketing Analytics, Network Analytics, Energy Analytics, Logistics & Supply Chain Analytics, Data Visualisation

**University College London** – BSc in Computer Science (2:1, Grade 68%) Sep 2021 - Jul 2024  
*Modules:* Machine Learning, Reinforcement Learning, Mathematics & Statistics, Algorithms, OOP, Software Engineering, Security

## SKILLS

|                        |                                                                                                                    |
|------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>     | Python, SQL, R, JavaScript, TypeScript, React, SvelteKit, HTML, TailwindCSS (Proficient); Java, Haskell, C (Basic) |
| <b>ML Libraries</b>    | pandas, OpenAI, pydantic, scikit-learn, XGBoost, PyTorch, FastText, SHAP, statsmodels, PyMuPDF, NetworkX           |
| <b>Technical Tools</b> | Git, AWS, FastAPI, Docker, Spark, PostgreSQL, Weaviate, RAG, Power BI, VWO, Google Analytics, Docs API             |
| <b>Finance</b>         | Mathematical Modelling, Forecasting, Portfolio Optimisation, Monte Carlo Simulation, CVar, Hedging                 |
| <b>Languages</b>       | English (Native), Cantonese (Fluent), Mandarin (Intermediate), Malay (Intermediate)                                |

## WORK EXPERIENCE

**Tecno International (Capstone Project, Imperial College London)** London, UK  
LLM Agent Engineer Jun 2025 - Aug 2025

- Designed multi-agent pipeline (Self-Assessment, Research, Strategy) using GPT-4.1 to automate SME ESG consulting
- Prototyped Weaviate RAG with hybrid BM25 + embeddings + Reciprocal Rank Fusion to retrieve and contextualise ESG insights
- Auto-generated ESG reports and enabled live in-chat edits, cutting analysis time 75% and freeing consultants for higher-order work
- Optimised token usage via context compression and caching, reducing API costs by 20% without quality loss
- Developed PDF ingestion pipeline using PyMuPDF + LLM extraction to auto-populate client profiles, cutting onboarding by 70%

**Anavio Capital Hedge Fund** London, UK  
Equity Analytics Intern – Long/Short Strategy Jul 2023 - Sep 2023

- Built a Python dashboard to uncover patterns in firm-level efficiency across 15+ metrics, identifying outliers for investment decisions
- Scaled coverage from 10 to 200 tickers daily, automated 80% of manual analysis, accelerating ideation for portfolio managers
- Enhanced pipeline robustness with 100+ Pytest tests and six error-handling decorators, minimising production risk and downtime
- Translated business rules into flowcharts to clarify logic and align tool design with non-technical stakeholders, halving delivery time

**GSE-M Technology Sdn. Bhd** Shah Alam, Malaysia  
Enterprise Resource Planning (ERP) Assistant May 2023 - Jul 2023

- Partnered with five departments to gather 30+ requirements, resolve critical bugs, and address operational pain points
- Streamlined ERP with consultants and co-led training for 90 staff, improving process efficiency by 25%

**British Red Cross** London, UK  
Data Analyst Sep 2022 - Apr 2023

- Developed 20+ Power BI dashboards to visualise behavioural trends in complaints across the UK for operational and senior teams
- Defined resolution KPIs and identified delay drivers, reducing resolution time by 50% and improving service outcomes for customers
- Built an ETL pipeline from four sources and used DAX for geographical drill-downs, enabling scalable and customised reporting
- Secured future development by presenting a streamlined issue resolution system along with a deployment manual to the Head of Data

**Subplace (B2C Subscription E-commerce Platform)** Kuala Lumpur, Malaysia  
Software Reliability Intern Jun 2022 - Aug 2022

- Cut maintenance time by 60% via modular and systematic test case templates, improving stability for a SQL-backed platform
- Resolved 200+ issues and introduced agile, test-driven development practices to enhance reliability and support continuous delivery

PROJECTS

**Multi-Channel Attribution.** Analysed 3.3M contacts and 240K orders to compare catalogue vs. email campaign effectiveness with PostgreSQL and Python. Computed attribution models (first-, last-, linear-click), revealing catalogue’s six times higher conversion. Segmented customers into 125 RFM cells, estimated CLV, and recommended targeting strategies based on breakeven cost and ROI

**Portfolio A/B Testing & Analytics.** Ran A/B tests with VWO to measure impact of clickbait vs professional phrasing on my portfolio website’s “Download CV” CTA; the professional version saw a 70.3% higher CTR (p = 0.043, statistically significant). Drove traffic via social media, Selenium bots, and a targeted Google Ads campaign; tracked conversions and bounce rate using event-based analytics

**Text Analytics on Song Lyrics.** Modelled 40,000+ lyrics through LASSO regression with n-grams and FastText embeddings to explore song popularity drivers. Discovered eight genre-level trends in explicitness and thematic content. Leveraged PCA-filtering and cosine similarity on word vectors to quantify semantic shifts between artists’ first albums and subsequent works

**RecSys Purchase Prediction.** Built a two-stage pipeline on 33M clicks and 1.1M purchases to predict purchasing sessions with XGBoost and forecast items bought via heuristic ranking. Engineered 21 features with temporal encodings and rarity filters. Tuned thresholds to optimise a Jaccard-based score (10,491) and used SHAP to interpret key drivers like session duration and item popularity

**Beat The Bookies.** Engineered a Premier League match predictor using 18 features across five seasons (older data degraded performance), achieving 51% validation and ~40% test accuracy with CatBoost. Feature-engineered Pi-rating variants with tuned hyperparameters and selected key variables using SHAP. Explored pre-game sentiment as a predictive signal using LLMs and Nitter API

**RIE Covariance Cleaning.** Developed a spectral cleaning algorithm using Rotationally Invariant Estimators, shrinking noise in high-dimensional lagged covariance matrices. Achieved 100% noise reduction in AR(0) and ~20% signal-to-noise improvement in AR(1), with more gains as dimensionality increased. Applicable to portfolio risk modelling, time series forecasting, and signal processing

**Logistics & Supply Chain Optimisation.** Simulated a multi-region supply chain implementing hybrid demand forecasting (SARIMA, Croston’s method) and cost-benefit analysis to guide factory and warehouse expansion. Applied adjusted reorder points and Silver-Meal heuristics to manage seasonal inventory, securing 2nd place in Imperial’s cohort

**Energy Demand Forecasting.** Developed and tuned XGBoost model to forecast UK electricity demand, engineering features like Heating Degree Days, holiday dummies, and temporal lags. Achieved 96.3% R<sup>2</sup>; generalises well with low residual autocorrelation

**Gas Plant Risk & Hedging.** Simulated 10,000 copula-based gas and electricity prices to assess plant risk and profitability. Designed a hedge (long electricity, short gas) that cut 95% Expected Shortfall by 29%, factoring in outages and cross-commodity correlations

**Gas Infrastructure Valuation with Real Options.** Valued a two-stage gas project using NPV and real options. Identified a break-even gas price of \$0.136/m<sup>3</sup> and option values of \$1.25M (start-of-year) and \$1.15M (mid-year), highlighting the effect of timing on valuation. Deferring Stage 2 added \$1.25M in value, demonstrating the financial benefits of flexibility under uncertainty

**TetrisAI.** Implemented a Tetris heuristics evaluation AI, scored 28,000+ points across 400 blocks, 80th percentile in UCL’s cohort

AWARDS AND CERTIFICATIONS

|                                                                                        |          |
|----------------------------------------------------------------------------------------|----------|
| DeepLearning.AI – Retrieval Augmented Generation (RAG)                                 | Aug 2025 |
| AmplifyME – Mergers and Acquisition Simulation with UBS                                | Feb 2025 |
| Boston Consulting Group – Introduction to Strategy Consulting (Forage)                 | Oct 2024 |
| Goldman Sachs – Software Engineering Experience Program (Forage)                       | Jul 2024 |
| Harvard University – CS50x Puzzle Day: Perfect Score (Top 13% Worldwide, Teams of Six) | Apr 2020 |
| Land Rover 4x4 In Schools Challenge – National Champions                               | Nov 2019 |
| Global Space Design Challenge – Innovation Award                                       | Jul 2019 |